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Logistic Regression

Contents

Logistic regression is similar to regression: the purpose is to build a regression equation to predict a dependent variable. However, there are some fundamental differences. First, logistic regression is based on maximum likelihood estimation, rather than ordinary least squares estimation. Additionally, logistic regression was designed to accommodate dependent variables measured on a nominal scale[1], a scale ordinary least squares multiple regression cannot handle (but see later in the chapter). Furthermore, logistic regression predictor results tend to be described in the context of odds/ratios, rather than beta-weights. Finally, there is also no obvious (or true) coefficient of determination (model R^2) in the context of logistic regression, although there are approximations.

In the simplest logistic regression case, i.e., the bivariate binary logistic regression, the dependent variable is measured on a dichotomous nominal scale and there is a single predictor variable. Bivariate implies two variables and binary implies a dependent variable measured dichotomously (e.g., continue/quit, default/no default). More complex logistic regression models can be tested with dependent variables measured on a nominal scale with more than

[1] However, versions have been developed to handle dependent variables measured on an ordinal scale.

two levels (i.e., multinomial logistic regression). Furthermore, logistic regression can also be performed with dependent variables measured with an ordinal scale (i.e., ordinal logistic regression). As per ordinary regression, logistic regression can be performed with one (bivariate) or more (multiple) predictor variables measured on any scale (e.g., dichotomous, continuous, ordinal).

Because logistic regression is based on odds/ratios, the interpretations of the results are, arguably, a little more complicated than ordinary regression results. Furthermore, there are several elements of the statistical results that are unique to logistic regression. This chapter assumes the reader has a decent command of bivariate and multiple regression. However, the chapter also starts things slowly. Specifically, the chapter begins with the simplest type of logistic regression, i.e., bivariate logistic regression with a dichotomously scored dependent variable. The example data set I analyzed in the chapter devoted to categorical data analysis (Chapter 4) will also be used here, i.e., the handedness and dyslexia study data. I then move onto another example with two predictor variables, which opens up the possibility of estimating unique effects, as well as interactions, as per multiple regression.

Bivariate Logistic Regression

In the chapter devoted to categorical data analysis, I presented simulated data on handedness and dyslexia (**Data File: handedness_dyslexia**). To refresh your memory, the total sample size of 250 adults reported their handedness (0 = right; 1 = left) and whether they were diagnosed with dyslexia (0 = no; 1 = yes). Based on a Pearson chi-square analysis (i.e., contingency table analysis), the null hypothesis of equal proportions of left-handers across the two dyslexia conditions (i.e., 2.2% versus 22.7%) was rejected, $\chi^2(1) = 22.03$, $p < .001$. Furthermore, *phi* (which is just a Pearson correlation on two dichotomously scored variables) was estimated at .297, which implied that 8.8% of the variance in dyslexia diagnosis was accounted for by handedness. Furthermore, because the manner in which the variables were coded, the positive *phi* value implied that left-handedness was associated with a greater chance of a dyslexia diagnosis. In fact, the percentage of non-dyslexic and dyslexic people who were left-handers was 2.2% and 22.7%, respectively. Such a result was essentially consistent with the results reported by Geschwind and Behan (1982).

Instead of a Pearson chi-square analysis, the handedness and dyslexia data could be analysed with bivariate logistic regression, because the dependent variable was measured on a dichotomous scale. Furthermore, logistic regression can handle independent variables scored on any level (interval/ratio, ordinal, and nominal). Although logistic regression is related closely to multiple regression, the hypotheses and results are framed in a slightly different way. Specifically, because the dependent variable is categorical, researchers typically speak about probabilities, that is, the probability of a case being classified into a particular group, on the basis of the independent variable scores. Another way to think about the probabilities associated with a bivariate regression analysis is to think about odds; specifically, odds/ratios.

Odds/ratios were covered in the Advanced Topics of Chapter 4 of this textbook. In the relevant section of chapter 4, I wrote that when there is the complete absence of an association between the two variables, the odds/ratio will equal 1.0. Thus, as the odds/ratio deviates from 1.0 (either lower or higher), the larger the magnitude of the effect. An intuitive formula for the calculation of an odds/ratio is:

$$\mathrm{OR} = \frac{a / b}{c / d} \tag{1}$$

Where a, b, c, and d refer to the frequencies associated with each of the cells within the contingency table analysis. What exactly constitutes a, b, c, and d is to some degree arbitrary, in the sense that a and c can certainly be interchanged, so long as b and d are also interchanged. In most cases, you will likely want to speak of the presence of the outcome in one group relative to the other. Thus, in the handedness and dyslexia example, it makes most sense to me to speak of the presence of dyslexia in left-handers relative to right-handers. Within the context of odds/ratios, the hypotheses are:

Hypotheses

Null hypothesis (H_0): The odds of being a male with dyslexia will be equal to being a female with dyslexia (i.e., exponentiation odds/ratio = 1.0).

Alternative hypothesis (H_1): The odds of being a male with dyslexia will be unequal to being a female with dyslexia (i.e., exponentiation odds/ratio ≈ 1.0).

Based on the current example's data, the observed frequencies were a = 5, b = 17, c = 5 and d = 223 (Watch Video 17.1: Bivariate Logistic Regression in SPSS).

handedness * dyslexia Crosstabulation

Count

		dyslexia		
		no	yes	Total
handedness	right	223	5	228
	left	17	5	22
Total		240	10	250

As can be seen below, the odds/ratio (OR) was estimated at 13.12. The value of 13.12 suggests that left-handers were 13.12 times more likely to be diagnosed with dyslexia than right-handers.

$$\mathrm{OR} = \frac{5/17}{5/223} = \frac{.294117}{.022422} = 13.12$$

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Although there are similarities between the Pearson 2x2 chi-square analysis and an odds/ratio analysis, the methods used to calculate the statistical significance of an odds/ratio is not the same as the Pearson chi-square analysis. Instead, researchers typically use a Wald statistic. Based on the results derived from SPSS, the Wald statistic was estimated at 14.30 and with 1 degree of freedom was significant statistically, $p < .001$.

Variables in the Equation

		B	S.E.	Wald	df	Sig.	Exp(B)
Step 1[a]	handedness	2.574	.681	14.30	1	.000	13.118
	Constant	-3.798	.452	70.53	1	.000	.022

a. Variable(s) entered on step 1: handedness.

You can see that SPSS labeled the odds/ratio as Exp(B), which is the exponentiation of the B (unstandardized) beta coefficient. The unstandardized beta-weight is essentially an odds/ratio, too, however, it is scaled in log-odds units, a more difficult scale to interpret. As described further below in the multiple logistic regression example, it is possible to obtain standardized beta-coefficients to allow for numerical comparisons between predictors.

Just like a bivariate regression, there is also an intercept in a logistic regression. It represents the value of Y when X is zero. As per the unstandardized beta-weight, however, the intercept is reported in log-odds units, which, again, is not readily interpretable. In this example, the intercept was found to be significant statistically, based on the Wald statistic, 70.53, $p < .001$. Thus, the intercept value of -3.798 is statistically significantly less than zero. As per regression, whether the intercept deviates from zero is not typically an interesting hypothesis to test.

In addition to the odds/ratio (i.e., 13.118), one may consider interpreting a measure of percentage of variance accounted for by the logistic regression model. In this context, SPSS reports two quasi-R^2 values: the Cox and Snell R^2 and the Nagelkerke R^2.

Model Summary

Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	71.671[a]	.048	.168

a. Estimation terminated at iteration number 6 because parameter estimates changed by less than .001.

As can be seen in the table entitled 'Model Summary', the Cox and Snell R^2 suggested that 4.8% of the variance in the dyslexia was accounted for by handedness; by comparison, the Nagelkerke R^2 suggested that 16.8% of the variance was accounted for. The Cox and Snell R^2 will always be smaller than the Nagelkerke R^2, sometimes substantially so, as in this example. Why? Well, it is well-established that a correlation with a variable measured on a dichotomous scale will not necessarily have a theoretical maximum value of 1.0 (Lord & Novick, 1968; Shrout & Parides, 1992). For example, a point-biserial correlation between a normally distributed continuous variable and a dichotomous variable with equal frequency of scores (i.e., 50% 0s and 50% 1s) cannot exceed .798 (Gradstein, 1986) (Watch Video 17.2: What is the maximum point-biserial correlation?). The *phi* coefficient, i.e., the correlation between two dichotomously scored variables, suffers from the same problem (Davenport & El-Sanhurry, 1991). For this reason, Nagelkerke (1991) developed an adjustment (max-rescaled R^2) to the Cox and Snell R^2 to essentially place the coefficient of determination on a theoretical scale from .00 to 1.0. However, when the continuously scored data are non-normally distributed, things become more complicated, and the theoretical maximum correlation between continuously and dichotomously scored variables can often exceed .798, and, in some scenarios, even reach 1.0 (Cheng & Liu, 2016). Arguably, this fact complicates the interpretation of the Nagelkerke R^2 adjustment, when the data are not perfectly normally distributed. Clearly, this is a thorny issue, one for which there is no obvious solution. In practice, I would report both Cox and Snell R^2 and Nagelkerke R^2 in a logistic regression study. However, I would not place much interpretive emphasis on either of them, at least not from the perspective of an R^2 value. Ultimately, they are very 'quasi' indicators of R^2. Consider that the Cox and Snell R^2 is really just a representation of the degree to which all of the predictor variable beta-weights are different from zero, as estimated via maximum likelihood (Cox & Snell, 1989). Consequently, all things considered, I would focus on the capacity of the model to classify cases correctly, as I demonstrate and discuss further below in the multiple logistic regression example. I also discuss a newer quasi-R^2 indicator, Tjur's R^2, which may be argued to have qualities more attractive than the more well-known Cox and Snell R^2 and Nagelkerke R^2.

Finally, I note that when I regressed dyslexia onto handedness with an "ordinary" (least squares) bivariate regression, I obtained a model R^2 = .084, which implied that 8.4% of the variance in dyslexia diagnosis was accounted for by handedness. Note that the model R^2 of .084 corresponds to phi^2 (recall, *phi* = .297). Additionally, the ordinary least squares regression R^2 corresponded more closely to the Cox and Snell R^2 than the Nagelkerke R^2. The problem with applying ordinary least squares regression to dichotomously scored dependent variables isn't that the beta-weights and model R^2 estimate are inaccurate. It's the accuracy of the standard errors and *p*-values that is of concern. In contrast to ordinary least squares regression, logistic regression is based on the natural logarithm of an odds ratio, which is more appropriate for inferring results from a sample to a population (i.e., more accurate standard errors), when the dependent variable is measured dichotomously.

Ultimately, the genuine benefits of logistic regression reveal themselves in a multiple logistic regression context, with two or more independent variables measured on a mixture of scales (e.g., interval ratio and nominal, for example). I demonstrate multiple logistic regression in the next example. Several new terms will be introduced, including the Lemeshow and Hosmer statistic, predicted probabilities, as well as a method to estimate standardized logistic regression beta-weights.

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Classification Accuracy

Although quasi-R^2 indices can offer some insights into the predictive capacity of a bivariate logistic regression equation, in my view, they lack the intuitive appeal associated with simply examining the percentage of cases classified by logistic regression equation correctly. Such a percentage is known as the 'crude hit rate'. Thus, for this example, one may wonder, 'What percentage of the non-dyslexics and dyslexics in the sample were classified correctly as non-dyslexic and dyslexic?' When the dependent variable is associated with a dichotomous scale, and there is an equal number of people who scored one value and the other, it is important to keep in mind that one could classify correctly 50% of the cases, on the basis of chance alone. Consequently, the null hypothesis is not 0% correct classification. The null hypothesis is 50% correct classification, when there is an equal number of cases who scored one value and the other value. When there is an unequal number of cases who scored one value over the other with respect to the dichotomous dependent variable, the null expectation is the mean associated with the dependent variable.

In SPSS, the first classification rate table is based on the absence of any predictors. Thus, it is the null hypothesis crude hit rate, which is based exclusively upon the mean of the dependent variable ('Constant is included in the model'). As can be seen in the SPSS table below entitled 'Classification Table[a,b]', the null hypothesis crude hit rate was found to be 96.0%. Thus, 96.0% of the cases were classified correctly as non-dyslexic and dyslexic – without any predictor(s). Consequently, a logistic regression model with one or more predictors needs to beat 96.0% correct classification.

As can be seen in the SPSS table below entitled 'Classification Table[a]' (presented below the first Classification Table), the crude hit rate remained 96.0%. Such a high percentage suggests that the logistic regression equation with the handedness predictor did a remarkable job at classifying the non-dyslexics from the dyslexics. However, such an impression would be based on a false premise, as the null model that included only the intercept (dependent variable mean[2]) classified correctly 96.0% of the cases. Furthermore, it is important to consider that there were only 10 dyslexia cases in the sample of 250, and the model which included handedness as a predictor classified *everyone* as non-dyslexic, 240 + 10 = 250. Thus, by predicting everyone as non-dyslexic, the model achieved 96.0% accuracy. This should not be regarded as impressive.

[2] The mean associated with dyslexia was .04. You need to subtract this value from 1 to obtain the null model crude hit rate (1-.04 = .96). The null model crude hit rate is always larger than .50, when an unequal number of cases scored 0 or 1 on the dependent variable.

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Classification Table[a,b]

	Observed		Predicted: dyslexia no	Predicted: dyslexia yes	Percentage Correct
Step 0	dyslexia	no	240	0	100.0
		yes	10	0	.0
	Overall Percentage				96.0

a. Constant is included in the model.

b. The cut value is .500

Classification Table[a]

	Observed		Predicted: dyslexia no	Predicted: dyslexia yes	Percentage Correct
Step 1	dyslexia	no	240	0	100.0
		yes	10	0	.0
	Overall Percentage				96.0

a. The cut value is .500

The fact that the model crude hit rate was not larger than the null model crud hit rate is inconsistent with the fact that the logistic regression analysis identified handedness as a statistically significant contributor to the logistic regression equation with dyslexia as the dependent variable. It is important to note, however, that several limitations associated with the crude hit rate have been articulated (Weiss, 1986). In particular, researchers have been recommended to not rely solely upon the crude hit rate to evaluate the predictive power of a logistic regression equation model. Instead, additional indices, such as sensitivity, specificity and Youden's index, may be regarded as insightful to evaluate the predictive capacity of a logistic regression equation.

Sensitivity and Specificity

Another way to think about results relevant to the prediction of a dichotomous outcome variable is 'sensitivity' and 'specificity'. In simple terms, you can think of sensitivity as the proportion of cases classified accurately by the model to have the characteristic of interest, where '0' is the absence of the characteristic and '1' is the presence of the characteristic. Sensitivity is also known as the "true positive rate". In the handedness and dyslexia example reported above, the sensitivity of handedness as a correct classifier of dyslexia was 0% (Watch Video 17.3: Sensitivity and Specificity Calculations). That's bad, because none of the dyslexic cases were predicted to be dyslexic accurately (0 / 10 = .00). In fact, the model did not even suggest any cases to be dyslexic at all.

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In contrast to sensitivity, you can think of specificity as the proportion of cases classified accurately to not have the characteristic of interest, again, where '0' is the absence of the characteristic and '1' is the presence of the characteristic. Sensitivity is also known as the "true negative rate". In the handedness and dyslexia example, the specificity of handedness as a correct classifier of non-dyslexia was 100%. At least on the surface, that's good, because it implies that all of the non-dyslexic cases were predicted accurately as non-dyslexic (240 / 240 = 1.0).

In an impressive sensitivity and specificity analysis, both sensitivity and specificity will be high, as such a value would suggest that the model is equally good at predicting the possession of the characteristic as not having the characteristic. Thus, in this handedness and dyslexia example, the results associated with the sensitivity and specificity analysis are not impressive, as the sensitivity was so low (i.e., 0%).

Of course, which ever level of the variable is coded '0' and '1' is usually arbitrary. Consequently, sensitivity and specificity values can be flipped, if the coding is reversed. Interestingly, in this example, the bivariate logistic regression model found handedness to be a statistically significant predictor of dyslexia. Additionally, the odds/ratio associated with handedness suggested that left-handers were more likely to be dyslexic than right-handers. However, as far as the prediction of individual cases is concerned, the model's results are not strong enough to yield respectable sensitivity and specificity values. This should be a concern for anyone who may wish to use a logistic regression equation for the purposes of prediction. Ultimately, the Cox and Snell R^2 was only about 5%, so, it should not really come as a surprise that the model is failing to do a good job of predicting who has dyslexia and who doesn't, on the basis of their handedness.

Youden's Index

An alternative approach to the evaluation of the predictive accuracy of a model is Youden's index (a.k.a., Youden's *J* statistic). The objective of Youden's index is to provide combined information about a model's capacity to discriminate between the two categories of the outcome variable (Youden, 1950). Youden's index values can theoretically range between 0 and 1, with a value closer to 1 indicative of a high level of accurate classification. The formula for Youden's index is simple, if sensitivity and specificity have been calculated, already:

$$J = \text{sensitivity} + \text{specificity} - 1 \qquad (2)$$

In the current handedness and dyslexia example, Youden's index was estimated at .00 (i.e., .00 + 1.0 – 1 = .00). Thus, the model that included handedness as a predictor of dyslexia was very poor.

Naturally, a limitation of Youden's index is that it does not provide information about the direction of misclassification. In practice, it can be of consequence whether a logistic regression equation produces more false positives than false negatives. For example, for some

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health related conditions, it may be a terrible outcome to miss the presence of a serious disease, particularly if the treatment is not particularly invasive or dangerous. By contrast, in other cases, it may be a terrible outcome to apply a treatment unnecessarily to deal with only a moderate condition, if the treatment has serious consequences.

Multiple Logistic Regression

The only difference between bivariate logistic regression and multiple logistic regression is that multiple logistic regression includes two or more independent variables. In a manner similar to multiple regression, the primary purpose of multiple logistic regression is to build a regression equation to predict maximally the dependent variable, or, more specifically, to classify cases into their groups accurately. Additionally, a researcher may want to evaluate which of the predictors are statistically significant unique contributors to the model, as per multiple regression.

Most of the terms associated with conducting a multiple logistic regression are the same as those mentioned above with respect to bivariate logistic regression. The main difference is that the beta-weights and odd/ratios represent unique effects, a valuable feature, assuming the predictor variables are inter-correlated. In addition to multiple logistic regression, in this section of the chapter, I demonstrate how to calculate predicted probabilities. I also show how to convert unstandardized beta-weights into standardized beta-weights to facilitate numerical comparisons between the predictors.

Morrow, McElroy, and Laczniak (1999) were interested in predicting employee turnover at an insurance company. In particular, Morrow et al. (1999) were interested in employee absenteeism and employee performance as predictors of employee turnover. Turnover was measured dichotomously (0 = stayer; 1 = leaver). Absenteeism was measured in number of hours absent from work over a 6-month period (ratio scale) and job performance was measured with manager ratings (ordinal scale). To research their question, Morrow et al. (1999) obtained data from 226 employees, half of which left the company and half stayed with the company. I have simulated data to correspond fairly closely to the results reported by Morrow et al. (1999) (**Data File: absenteeism_turnover**).

It is good practice to run Pearson correlations between your variables, if only to get a general sense of what is going on, as Pearson correlations are relatively easy to interpret.[3] As can be seen in the SPSS table entitled, ‘Correlations’, higher-levels of absenteeism were associated with a greater likelihood of leaving the company, based on a point-biserial (Pearson) correlation (r = .203, p = .002). Correspondingly, higher-levels of job performance were associated with a lesser likelihood of leaving the company, again, based on a point-

[3] This recommendation is made on the basis that the variables have been measured on a scale upon which it is sensible to apply Pearson correlations. A variable measured on a nominal scale with more than two levels is probably not appropriate to submit to a Pearson correlation analysis, for example.

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biserial (Pearson) correlation (r = -.290, p < .001). Finally, absenteeism and job performance were correlated negatively, based on a Pearson correlation (r = -.138, p = .038).[4] Given the pattern of correlations, it would be useful to conduct an analysis to predict turnover, in a manner similar to a multiple regression. However, in this case, the dependent variable was measured on a dichotomous scale; therefore, multiple logistic regression is, arguably, a more justifiable option to analyze these data.

Correlations

		turnover	absenteeism	performance
turnover	Pearson Correlation	1	.203**	-.290**
	Sig. (2-tailed)		.002	.000
	N	226	226	226
absenteeism	Pearson Correlation	.203**	1	-.138*
	Sig. (2-tailed)	.002		.038
	N	226	226	226
performance	Pearson Correlation	-.290**	-.138*	1
	Sig. (2-tailed)	.000	.038	
	N	226	226	226

**. Correlation is significant at the 0.01 level (2-tailed).

*. Correlation is significant at the 0.05 level (2-tailed).

Next, I conducted the multiple logistic regression in SPSS (Watch Video 17.4: Multiple Binary Logistic Regression in SPSS). The first couple of tables produced by SPSS are relevant to the sample size (and missing cases; 'Case Processing Summary'), as well as to the nature (levels; labels) of the dependent variable ('Dependent Variable Encoding'). Recall that, in this example, the dependent variable was scored dichotomously: 0 = stayer; 1 = leaver. Although I don't show all of the tables, here, SPSS reports the 'Classification Table' associated with 'Step 0', which is relevant to a model that does not include any predictors. It's the null model, essentially. It can be seen that the null model predicted 50.0% of the cases correctly, which you expect when 50% of the cases scored zero and 50% scored 1. The model which includes the predictors must exceed 50% accuracy, preferably by a substantial margin. It should be noted that null model correct classification can be substantially higher than 50%, in some cases, so it is important to consider the null model results.

[4] Morrow et al. (1999) actually reported a non-significant, positively directed correlation between absenteeism and performance (r = .06). I simulated the data to be a little more interesting and consistent with my expectations. ☺

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Classification Table[a,b]

Observed			Predicted: turnover stayer	Predicted: turnover leaver	Percentage Correct
Step 0	turnover	stayer	0	113	.0
		leaver	0	113	100.0
	Overall Percentage				50.0

a. Constant is included in the model.

b. The cut value is .500

Next, things become more interesting: The 'Block 1: Method = Enter' results are reported, which includes the predictors. First, the test of the statistical significance of the overall model is reported in a SPSS table entitled 'Omnibus Tests of Model Coefficients'. It can be seen the chi-square statistic with 2 degrees of freedom was significant statistically, $p < .001$. Thus, the combination of all of the predictors in the model achieved some statistically significant level of accuracy in predicting the dependent variable (turnover). As a general statement, you may view this chi-square test as similar to the ANOVA F-test reported for the overall Model R^2 in a multiple regression analysis. However, an argument can be made that the omnibus chi-square test is inappropriate for models with one or more continuously scored predictors variables (Hosmer, Hosmer, Le Cessie, & Lemeshow, 1997), as was the case, here.

Omnibus Tests of Model Coefficients

		Chi-square	df	Sig.
Step 1	Step	29.734	2	.000
	Block	29.734	2	.000
	Model	29.734	2	.000

Next, SPSS reported the quasi-percentages of variance accounted for by the model with the two predictors in a table entitled 'Omnibus Tests of Model Coefficients'. The Cox and Snell R^2 of .123 suggests that 12.3% of the variance in turnover was accounted for by the two predictors. By comparison, the Nagelkerke R^2 was estimated at .164, which suggests that 16.4% of the variance in turnover was accounted for by the two predictors. For a point of comparison, a regular (ordinary least squares) multiple regression yielded a Model R^2 of .111, which is closer to the Cox and Snell R^2 estimate than the Nagelkerke R^2 estimate, as expected. Although approximately 12% of the variance in turnover was accounted for by the model with two predictors, the results suggest that 88% of the variance was due to factors other than absenteeism and job performance. Evidently, whether an employee leaves an organization or not is more complicated than absenteeism and job performance.

Model Summary

Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	283.569[a]	.123	.164

a. Estimation terminated at iteration number 5 because parameter estimates changed by less than .001.

Next, SPSS reported the Hosmer and Lemeshow (H-L) test. The H-L test was developed to overcome the limitation of the chi-square omnibus test, which is restricted to the use with independent variables scored on a nominal (categorical) scale. The Hosmer and Lemeshow test can more appropriately handle models with one or more independent variables scored on a continuous scale (although it has been argued to lack statistical power in some plausible situations). Unlike most statistics, a non-significant *p*-value associated with the H-L test implies that null model of no predictive capacity can be *rejected*. Thus, in this example, the H-L test chi-square statistic of 7.343, with 8 degrees freedom, was non-significant, p = .500. That's a good thing, if you were hoping to argue in favor of the usefulness of the model with two predictors (both of which were scored on continuous scales).

Hosmer and Lemeshow Test

Step	Chi-square	df	Sig.
1	7.343	8	.500

Next, SPSS reported the 'Contingency Table associated with the Hosmer and Lemeshow Test'. You can see that there are columns of observed and expected frequencies. The H-L test is a chi-square test of the difference between the observed and expected cell frequencies. A statistically significant lack of fit between the observed and expected (model predicted) cell frequencies implies that the model does not adequately predict the data. Thus, a *p*-value less than .05 is a "bad" thing with respect to evidence of a statistically significant (useful) logistic regression model. In practice, what you also want to see, here, is that the H-L test was able to create more than 5 steps (or groups or deciles). In this case, 10 steps/groups (or ten deciles) were created, which is the maximum the H-L test will create. These additional results help support the validity of the non-significant H-L test: the model appears to have predicted turnover in a statistically significant way.

Contingency Table for Hosmer and Lemeshow Test

		turnover = stayer		turnover = leaver		
		Observed	Expected	Observed	Expected	Total
Step 1	1	23	19.690	0	3.310	23
	2	12	15.754	11	7.246	23
	3	13	13.337	10	9.663	23
	4	11	11.956	12	11.044	23
	5	11	10.764	12	12.236	23
	6	11	10.211	12	12.789	23
	7	9	9.888	14	13.112	23
	8	10	9.429	13	13.571	23
	9	9	8.200	14	14.800	23
	10	4	3.771	15	15.229	19

Next, SPSS reported very important results in a SPSS table entitled 'Classification Table'. It can be seen that 61.9% of the cases in the sample were classified correctly by the model. Although such a percentage may seem high in an absolute sense, keep in mind that the null model classified correctly 50% of the cases. Arguably, one should interpret a numerical difference between the null model and predictor model correct classification rates as significant statistically only if the H-L statistic is non-significant (again, a H-L $p > .05$ implies a model that is predicting the dependent variable in manner that is sufficiently beyond chance to declare as significant statistically).

Classification Table[a]

			Predicted		
			turnover		Percentage Correct
Observed			stayer	leaver	
Step 1	turnover	stayer	59	54	52.2
		leaver	32	81	71.7
	Overall Percentage				61.9

a. The cut value is .500

Finally, SPSS reported the key results associated with the regression equation in a table entitled 'Variables in the Equation'. First, SPSS refers to the intercept as the 'Constant', which in this example corresponded to 3.689. Recall that an intercept in multiple regression is the value of *Y* when all *X*s are zero. Only in rare cases does the intercept provide insightful information; however, it is necessary to have the intercept to use the regression equation. In this example, the intercept was observed to be significant statistically, based on the Wald statistic, which follows the chi-square distribution, $\chi^2(1) = 11.65$, $p = .001$. Thus, the intercept may be regarded as statistically significantly different from zero.

Next, the unstandardized beta weights associated with two predictors are considered. It can be seen in the table entitled 'Variables in the Equation' that both absenteeism, $B = .015$, $\chi^2(1)$, $p = .014$, and job performance, $B = -.631$, $\chi^2(1)$, $p < .001$, were both statistically significant contributors to the multiple logistic regression equation. As per multiple regression, these beta-weights may be considered unique effects, i.e., controlling for the effects of the other independent variable(s). The direction of the beta-weights suggest that higher levels of absenteeism were associated with a greater probability of leaving; by contrast, higher levels of job performance were associated with a lesser probability of leaving.

Variables in the Equation

		B	S.E.	Wald	df	Sig.	Exp(B)	95% C.I.for EXP(B)	
								Lower	Upper
Step 1[a]	absenteeism	.015	.006	6.000	1	.014	1.016	1.003	1.028
	performance	-.631	.165	14.61	1	.000	.532	.385	.735
	Constant	3.689	1.081	11.65	1	.001	39.995		

a. Variable(s) entered on step 1: absenteeism, performance.

Because absenteeism and job performance were measured on different scales, it is difficult to suggest which of the two independent variables may have contributed more substantially to the regression equation; however, the magnitude of the Wald statistics do give some suggestion that job performance was a more substantial unique contributor to the model. The odds/ratios also give some suggestion that job performance was a more important unique predictor of turnover. Absenteeism was associated with an odds/ratio of 1.016, which implies that a 1 unit increase in absenteeism was associated with a 1.6% *greater* chance of leaving the company, independently of the effects of job performance. I obtained the 1.6% by subtracting the value of 1 from the odds/ratio and then multiplying the residual by 100: (1.016 – 1) * 100 = 1.6%. For odds/ratios less than 1.0, the process is a slightly different. One needs to subtract the odds/ratio from 1 and then multiply the residual by 100. Thus, in this example, a 1 unit increase in job performance was associated with a 46.8% (i.e., (1 - .532) * 100)) *lesser* chance of leaving, independently of the effects of absenteeism. It's a lesser chance, because the beta-weight was negative. The 'Variables in the Equation' table also includes the 95% confidence intervals associated with the odds/ratio point-estimates should you wish to report them.

When interpreting the results above, keep in mind that absenteeism and job performance were associated with the following standard deviations of 27.64 and 1.22, respectively. Thus, in reality, the one unit increases in absenteeism and job performance are not really equivalent, as the standard deviations are very different. In order to facilitate interpretations of the magnitude of effects across predictors, it is useful to obtain standardized beta-weights. Unfortunately, most programs do not report standardized beta-weights for logistic regression. However, with a few fairly easily implemented steps, standardized beta-weights can be obtained. The first step involves calculating predicted probabilities, as I demonstrate next.

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Predicted Probabilities

In the context of logistic regression, a predicted probability is an estimate that corresponds to the group a case is predicted to belong (e.g., stayer = 0; or leaver = 1), on the basis of the logistic regression equation. In the context of binary logistic regression, predicted probabilities range in value from .0001 to .9999. Although predicted probabilities are continuous in nature, they can be recoded into dichotomous values via rounding.

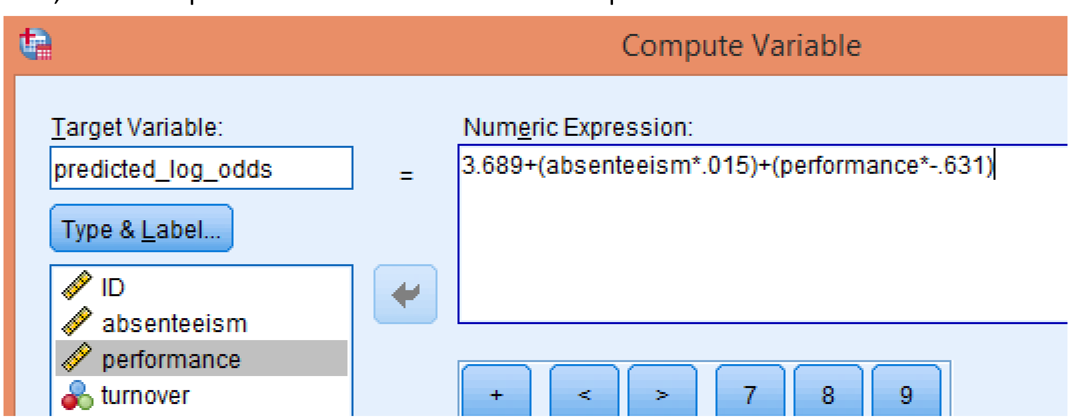
It is important to note that the logistic regression equation reported by SPSS is expressed on a log odds scale. Consequently, predicted values derived from the reported logistic regression equation will also be expressed on a log odds scale, which is not easy to interpret, nor does it produce predicted probabilities that are easy to interpret. Fortunately, the log odds scaled predicted values derived from the logistic regression equation can be converted fairly easily into adjusted predicted probabilities, which are easier to interpret. The process involves three steps: (1) convert the log odds results into an odds scale; (2) exponentiate the odds scaled values; and (3) convert the exponentiated odds scaled values into predicted probabilities ([Watch Video 17.5: Calculating Predicted Probabilities in SPSS]()).

The first step involves simply using the logistic regression equation as it is reported by SPSS. In this example, that would involve the following equation:

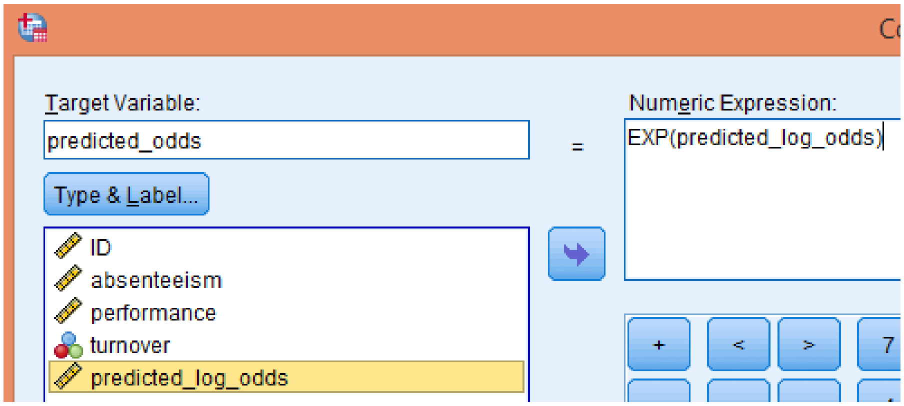
$$\text{Predicted log odds} = \text{intercept} + (X_i * \text{B}) \qquad (3)$$

Where the intercept corresponds to the 'Constant' in the SPSS table entitled 'Variables in the Equation' and B corresponds to the unstandardized beta-weight in the same table. The X_i symbol corresponds to the any independent variable value (or all of them).

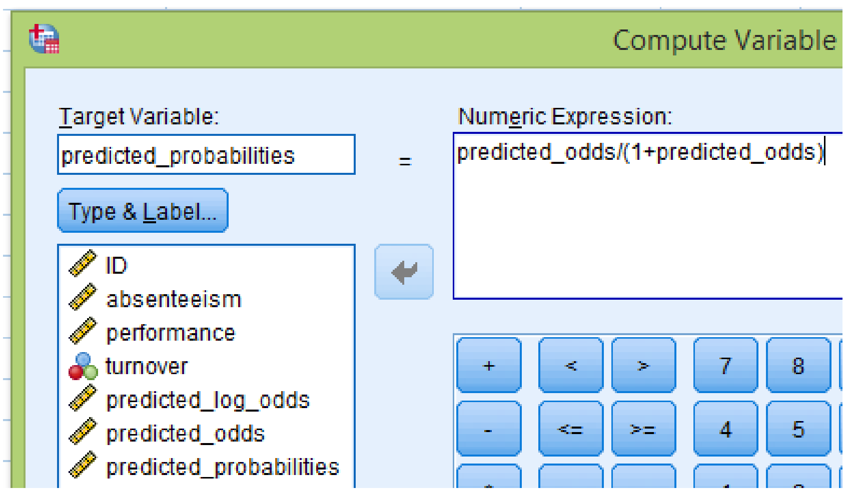
In SPSS, such an equation can be solved with the Compute Variable function:



The above function will produce a new variable in SPSS called 'predicted_log_odds'. Once the predicted log odds are calculated, they need to be exponentiated (step 2), again, through the Compute Variable function:



Finally, create the predicted probabilities:



You should see that SPSS has created a new variable called 'predicted_probabilities', with values ranging from .01 to .93. Predicted probability values less than .50 suggest that the employee stayed with the company (recall that 0 = stayer). By contrast, predicted probabilities greater than .50 suggest that the employee left the company (recall that 1 = leaver). Correspondingly, you can round down and up the predicted probabilities to take on values of 0 and 1, respectively, which corresponds to the logistic regression equation's prediction of group membership. If you do so, you'll find that the logistic regression equation predicted 91 cases (40.3%) as stayers and 135 cases (59.7%) as leavers. However, in actuality, exactly 50% of the cases were stayers and leavers. Furthermore, if you calculate the difference between the predicted group membership scores and the actual group membership scores, you'll find that 61.9% of the values were 0, which implies correct classification. Finally, when I calculated the Pearson correlation between the predicted group membership values and the actual group membership values, I obtained a *phi* = .244, which corresponds to phi^2 = .059. The value of .059 suggests that the logistic

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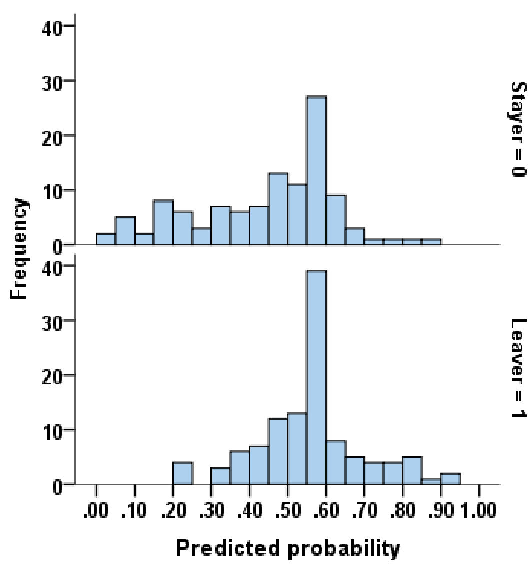
regression model accounted for 5.9% of the variance in the dependent variable. However, even the Cox and Snell R^2 was substantially larger than .059 (i.e., it was .123). Clearly, quasi-R^2 values, such as Cox and Snell R^2 and Nagelkerke R^2, are not R^2 values as we know and love them from the ordinary least squares framework (i.e., conventional multiple regression).

Tjur's Quasi-R^2

A fairly simple and attractive approach to the evaluation of the predictive capacity of a logistic regression model is to plot the predicted probabilities in two histograms: one histogram for observed scores equal to 0 and one histogram for observed scored equal to 1. This assumes the dependent variable has been scores 0 and 1, as it was in the turnover study example (stayers = 0; leavers = 1). If other values were specified, the principle would be the same. When I created such a graph with the turnover example data, I obtained the histogram below (Figure C13.1). It can be seen that there is a substantial amount of overlap across the two distribution. Such an observation implies that the logistic regression model has not predicted group membership particularly well. When a model works very well, you will see most of the predicted probabilities for one group on the left side and most of the predicted probabilities for the other group on the right side. Tjur (2009) suggested calculating the mean predicted probabilities associated with each group. Then, calculate the absolute difference between the two means. In this context, the absolute mean difference is known as the coefficient of discrimination or Tjur's R^2. In this example, the predicted probability means were .478 and .717 for the stayers and leavers groups, respectively. Thus, Tjur's R^2 = .239. Depending on your point of view, you would consider reporting such a value as opposed to Cox and Snell's R^2 or Nagelkerke's R^2.

Figure C13.1.



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Standardized Beta-Weights

In addition to calculating odd/ratios, predicted probabilities, and predicted group membership in a multiple logistic regression analysis, it can be useful to obtain standardized beta-weights. For example, standardized beta-weights allow for some numerical comparisons between predictors.[5] The interpretation of a logistic regression standardized beta-weight is similar to a standardized beta-weight in ordinary least squares multiple regression. Specifically, a one standard deviation increase in the predictor is associated with a unique increase (positive beta-weight) or decrease (negative beta-weight) in the dependent variable equal to one beta-weight of a standard deviation in the dependent variable, controlling for the effects of the other independent variables.

Several approaches to the standardization of logistic regression beta-weights have been proposed (Menard, 2004). Unfortunately, programs such as SPSS do not report standardized beta-weights in a multiple logistic regression. Fortunately, however, an attractive approach to the standardization of a beta-weight can be calculated with the following formula (Menard, 2004):

$$\beta = b * SD_{\text{IV}} * R \,/\, SD_{\text{Logit}Y} \tag{4}$$

Where b = unstandardized beta-weight, SD_{IV} = the standard deviation of the predictor variable scores, R = Cox and Snell R; and $SD_{\text{Logit}Y}$ = the standard deviation of the predicted log odds variable scores (Watch Video 17.6: Standardized Beta-Weights for Logistic Regression). Based on previous analyses reported above, the unstandardized beta-weights associated with the absenteeism and job performance predictor variables were -.631 and 3.689, respectively. Furthermore, the corresponding predictor variable standard deviations were 27.639 and 1.220, respectively. Additionally, the Cox and Snell R was equal to .351 ($\sqrt{.123}$). Finally, the standard deviation associated with the predicted log odds variable scores was $SD_{\text{Logit}Y}$ = .923. Therefore:

Job Performance β = -.631 * 1.220 * .351 / .923 = -.293
Absenteeism β = .015 * 27.639 * .351 / .923 = .158

Thus, a one standard deviation increase in job performance was associated with a -.293 of a standard deviation decrease in turnover, controlling for the effects of absenteeism. Correspondingly, a one standard deviation increase in absenteeism was associated with .158 of a standard deviation increase in turn over, controlling for the effects of job performance. Thus, based strictly upon a numerical comparison, job performance made a more substantial contribution to the regression equation. However, the numerical difference would need to be tested statistically, in order to reject the null hypothesis of no difference between the two standardized beta-weights. I'm not familiar with an easily implementable test of such a

[5] To make claims about the difference between two standardized beta-weights statistically would require a statistical test of the difference between the beta-weights.

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hypothesis. Finally, I note that the Cox and Snell *R* was used in the formula above to derive the standardized beta-weights. However, arguments could be made to use other measures of *R* (Hong & Ryu, 2006).

Logistic Regression: Assumptions

Like most any statistical analysis, logistic regression has assumptions. Naturally, random sampling is assumed, but almost always violated (the show must go on…). Unlike conventional multiple regression, logistic regression does not assume normally distributed independent or dependent variable sores. However, in a manner similar to multiple regression, there is a pervasive notion that researchers should have at least 10 participants per independent variable (e.g., Concato & Feinstein, 1997). More recent simulation research suggests that this rule of thumb cannot be trusted, as other factors, like total sample, should be considered (van Smeden et al., 2019). Given the close relationship between multiple logistic regression and multiple regression (Hellevik, 2009), my view is that researchers should be focusing on statistical power, when contemplating logistic regression sample size requirements. In this context, the contemporaneous analytical and simulation research on multiple regression has found that one should take into account the magnitude of the inter-associations between the predictor variables, when considering the sample sizes required to conduct a useful multiple logistic regression (see Maxwell, 2000). As a general statement, a sample size of approximately 100 would be required to conduct a logistic regression, if one is primarily interested in the overall amount of prediction made by the model (borrowing from Green, 1991). By contrast, if a researcher is interested in evaluating each predictor for its unique contribution to the logistic regression equation, larger sample sizes will be required (Maxwell, 2000).

Multicolinearity is also a potential consideration in the interpretation of multiple logistic regression results. The presence of substantial multicolinearity impacts the stability of the results, particularly the logistic regression beta-weights (and standard errors). Curiously, many statistical programs do not report tolerance or variance inflation factors for logistic regression results, but they do for multiple regression. Consequently, researchers can analyze their data within multiple regression, in order to get the tolerance and/or variance inflation factors. Each predictor variable should have a tolerance of .10 or greater and a variance inflation factor less than 10, in order to conduct the multiple logistic regression with confidence. See the chapter on multiple regression for further information on multicolinearity.

The data are assumed to be derived from a cross-sectional, non-repeated measures design (e.g., people respond to each measure only once). Finally, for binary logistic regression, it is assumed that the dependent variable is measured dichotomously. It's inappropriate to use a continuously scored dependent variable for logistic regression. For dependent variables with

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more than two (nominal) categories, multinomial logistic regression is an option, and for dependent variables measured on an ordinal scale, ordinal logistic regression is an option.

Ordinary Multiple Regression with a Dichotomously Scored Dependent Variable

Some have argued that critiques against using ordinary least squares multiple regression with a dichotomously scored dependent variable are not compelling. For example, Hellevik (2009) demonstrated analytically and empirically that ordinary least squares multiple regression cannot only work (i.e., provide essentially accurate results), but even provide more insightful and less misleading results. I would add to Hellevik's (2009) arguments that the application of bootstrapping to multiple regression with a dichotomously scored dependent variable may be especially attractive, as the standard errors, *p*-values, and confidence intervals will almost certainly be associated with relatively accurate standard errors and confidence intervals.

To help support the argument made, here, I conducted a multiple regression analysis with bootstrapping on the employment turnover study data. As can be seen in the SPSS tables below, the model with absenteeism and job performance as predictors yielded a model R^2 = .111. Furthermore, the model R^2 was significant statistically, $F(2, 223) = 13.952$, $p < .001$. Recall that the Cox and Snell quasi-R^2 was .123, which was very similar to the model R^2 of .111.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.333[a]	.111	.103	.4745

a. Predictors: (Constant), performance, absenteeism

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	6.284	2	3.142	13.952	.000[b]
	Residual	50.216	223	.225		
	Total	56.500	225			

a. Dependent Variable: turnover

b. Predictors: (Constant), performance, absenteeism

As can be seen in the SPSS table entitled 'Coefficients', the standardized beta-weights were .167 and -.267, which correspond closely to the standardized logistic regression beta-weights reported earlier.

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Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.
1	(Constant)	1.139	.183		6.229	.000
	absenteeism	.003	.001	.167	2.612	.010
	performance	-.110	.026	-.267	-4.186	.000

a. Dependent Variable: turnover

Finally, both predictors were found to be significant unique contributors to the regression equation, based on the bootstrapped *p*-values. Again, the *p*-value results were similar to those obtained from the multiple logistic regression results. Thus, Hellevik's (2009) position appears to have some merit. Whether you want to abandon logistic regression in favor of multiple regression may be controversial. However, running the analysis via multiple regression, prior to conducting the logistic regression, can be a good way to verify the accuracy of the standardized beta-weights (in rough terms) you calculated from the multiple logistic regression.

Bootstrap for Coefficients

Model		B	Bootstrap[a] Bias	Bootstrap[a] Std. Error	Bootstrap[a] Sig. (2-tailed)	BCa 95% Confidence Interval Lower	BCa 95% Confidence Interval Upper
1	(Constant)	1.139	.003	.165	.000	.824	1.488
	absenteeism	.003	.000	.001	.008	.001	.006
	performance	-.110	-.001	.021	.000	-.156	-.071

a. Unless otherwise noted, bootstrap results are based on 2000 bootstrap samples

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Suggested Reading

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